High Dimensionality

Summary sheet — Deep Learning (S.-A. Berrani, ENSIA 2026–2027)

Core idea & notation

High-dimensional data (dimension > 3) escapes our intuition: distances, volumes and sample requirements behave in counter-intuitive ways. All these difficulties are called the **dimensionality curse**. Analysis, exploration, processing and management all become harder.

- d — dimension (number of variables); n — number of individuals (samples).
- $V_d(r)$ — volume of a d-dimensional hyper-sphere of radius r; ε — width of the frame in the hyper-cube.
- $D_{\min}, D_{\max}$ — distance from a query point to its nearest / furthest neighbour.

1. The dimensionality curse

Optimization: difficulty of optimizing in high-dimensional spaces with *exhaustive* methods.
Statistics: to keep the same reliability of an estimate, the number of samples must grow *exponentially* with d.

Identification. More variables *and* more individuals ⇒ less intuition, costly processing, harder analysis. Whenever a method needs "enough points in every region of space", expect the curse.

1.1 Quantities that grow exponentially with d

To cover or explore all possible configurations, the required effort grows exponentially with dimension:

Quantity	Growth	Illustration
Binary combinations of d variables	2^d	Each new dimension doubles the number of states ($10 \to 10^3, 20 \to 10^6$)

2. Loss of the notion of distance

When d increases (under data uniformly distributed on all dimensions and independent):

- the distance to the **nearest** neighbour increases;
- the distance to the **furthest** neighbour increases;
- but $D_{\min}$ increases **faster** than $D_{\max}$.

Limit:

$$d \to \infty \implies \boxed{D_{\min} \approx D_{\max}} \qquad \left(\text{i.e. } \frac{D_{\max}}{D_{\min}} \to 1\right)$$

Consequence. Nearest and furthest neighbours become indistinguishable: all points look equally far apart, so distance-based methods (nearest neighbour, clustering with Euclidean distance) lose meaning.

3. Volume of the hyper-sphere

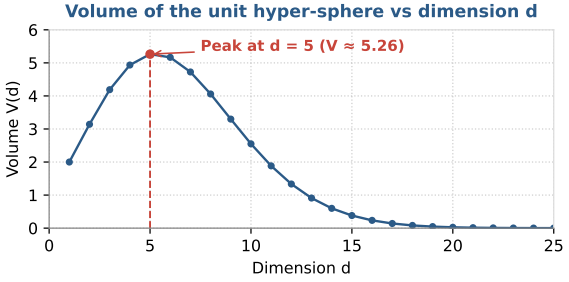
Formula:

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)} r^d$$

Behavior (unit sphere, $r = 1$):

- The volume initially increases, reaches its maximum at $d = 5$ ($V_5(1) \approx 5.26$), and then rapidly collapses to 0 as $d \to \infty$.
- Relative to the enclosing cube (side 2, volume 2^d), the sphere's volume ratio drops exponentially to 0:

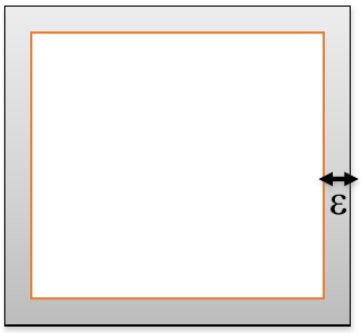
$$\lim_{d\to\infty} \frac{V_{\text{sphere}}}{V_{\text{cube}}} = 0$$



- *The volume of the cube concentrates in its corners.*

4. Exercise: hyper-cube and its frame

Setup: points uniformly distributed in the hyper-cube $[0,1]^d$. The **frame** (grey) is the set of points at distance $\leq \varepsilon$ from the edge, $0 < \varepsilon < \frac{1}{2}$. The remaining inner hyper-cube is $[\varepsilon, 1-\varepsilon]^d$, of side $1-2\varepsilon$.



Hyper-cube and ε-frame

Probabilities (volume ratios):

$$P(\text{inner}) = \frac{\text{Vol}([\varepsilon, 1-\varepsilon]^d)}{\text{Vol}([0,1]^d)} = (1-2\varepsilon)^d, \qquad \boxed{P(\text{frame}) = 1 - (1-2\varepsilon)^d}$$

Asymptotic limit ($d \to \infty$): Since $0 < 1-2\varepsilon < 1$, $(1-2\varepsilon)^d \xrightarrow{d\to\infty} 0$, hence:

$$\lim_{d\to\infty} P(\text{frame}) = 1$$

Example: $\varepsilon = 0.05$ (so $1-2\varepsilon = 0.9$):

$$d = 2 : 0.19, \quad d = 3 : 0.27, \quad d = 10 : 0.65, \quad d = 50 : 0.995, \quad d = 100 : 0.99997$$

Number of dimensions for $P(\text{frame}) \geq p$: $d \geq \frac{\ln(1-p)}{\ln(1-2\varepsilon)}$; for $p = 0.99$: $d \geq 43.7$, i.e. $\mathbf{d \geq 44}$.

What we notice. The probability curve increases and saturates at 1: in high dimension, **almost every point lies near the edge** (in the ε-frame) and the "centre" is empty. Data are sparse and isolated.

5. Solutions

	1. Reduce the variables	**2. Reduce the instances**
Idea	Create new variables from the original ones while losing as little information as possible	Partition the dataset by grouping similar individuals
Acts on	Columns (features), $p \to q < p$	Rows (individuals)
Example	**PCA** (Principal Component Analysis)	**Clustering** (unsupervised classification)